# Implicit CLIP Prior Decoupling for Few-Shot Remote Sensing Image Segmentation

Zhiyu Jiang, Ye Yuan, Dandan Ma, Qi Wang, *Senior Member, IEEE*, and Yuan Yuan, *Senior Member, IEEE*

***Abstract*—Few-shot segmentation (FSS) in remote sensing aims to achieve segmentation of novel categories in query images using limited annotated support images. Despite extensive research, the significant intraclass differences of remote sensing targets continue to hinder progress in this field. Pretrained vision–language models (VLMs) possess strong generalization capabilities, and their cross-modal information can effectively mitigate intraclass variance issues. However, VLMs rarely focus on dense prediction tasks, and the complexity of remote sensing imagery limits the effectiveness of existing attempts on FSS tasks. To address this issue, this article proposes an implicit contrastive language-image pretraining (CLIP) prior decoupling network (ICPD-Net), which mines effective cross-modal priors from VLMs and leverages ranking information to improve visual metric strategies. Specifically, the implicit prior decoupling module (IPDM) utilizes ambiguous foreground–background vision–language similarities to construct class-agnostic prompts, while employing a prior learner to mine implicit vision–language priors that alleviate intraclass differences. To fully leverage cross-modal information, the reliable feature fusion module (RFFM) utilizes vision–language priors to obtain high-confidence query features for fusion with support features and further mitigates intraclass differences through a self-support paradigm. Finally, the dual visual priors module (DVPM) introduces a novel rank information prior for visual feature measurement. This approach constructs an effective metric learning method by combining the ranking relationships of Euclidean distances between support-query features with the normalized discounted cumulative gain (NDCG) algorithm, while comprehensively exploring visual metric relationships through a traditional cosine similarity prior. Extensive experiments on iSAID-$5^i$ and DLRSD-$5^i$ demonstrate that our method achieves significant improvements. Particularly under the one-shot setting, our approach shows exceptional effectiveness, outperforming state-of-the-art methods by up to 11.48% on the iSAID-$5^i$ dataset. The code of the ICPD-Net is available at https://github.com/yeh15/ICPD-Net**

***Index Terms*—Cross-modal learning, few-shot segmentation (FSS), metric learning, remote sensing.**

## I. Introduction

REMOTE sensing semantic segmentation [1] is a crucial task in the understanding of remote sensing images, aiming to distinguish categories present in the images at the pixel level. This technique holds significant applications across numerous fields, including agricultural management [2], urban planning [3], disaster response [4], and more. With the advancement of deep learning, models [5] based on advanced architectures such as convolutional networks and self-attention networks have achieved remarkable performance in the semantic segmentation of remote sensing images. However, these methods typically require an adequate amount of densely annotated data. The complexity inherent in remote sensing tasks makes the collection and annotation of data challenging, thereby limiting the application of these methods. Furthermore, traditional methods, which typically require that the test data and the training data be drawn from the same distribution, are unable to rapidly adapt to novel categories and concepts.

To address these issues, few-shot segmentation (FSS) [9] has been proposed, which can achieve segmentation of novel category targets using only a handful of annotated samples and efficiently generalize to novel concepts. Existing approaches [10] to FSS generally rely on a handful of annotated samples, termed the support set, to furnish guidance for the unannotated query set. Consequently, these methodologies encounter issues including significant intraclass variation and suboptimal quality of guidance.

Intraclass variation is a key issue present in remote sensing FSS. In addition to variations in appearance and scale within object categories, remote sensing tasks must also contend with arbitrary rotations and incomplete target representations in images. These phenomena contribute to intraclass variations, thereby affecting the performance of FSS. To tackle this challenge, many studies have offered valuable solutions. For example, some methods [11], [12] introduce additional unlabeled images or prior information, some methods [13] adopt multiprototype guidance strategies to enhance the perception of feature diversity, and some methods [14], [15] leverage the self-support mechanism [16] by utilizing the information within the query image itself to guide its own segmentation. However, the limited annotated information remains insufficient to capture all variations within the categories. Recent studies [17] have found that the strong generalization performance of vision–language models (VLMs) like contrastive language-image pretraining (CLIP) [18] can effectively alleviate the issue of intraclass variation in few-shot classification. However, since the capabilities of VLMs are geared toward image-level tasks, they cannot be directly applied to dense prediction tasks. Some methods [6], [19] use GradCAM or directly employ feature cosine similarity for natural image segmentation tasks. Some methods [7], [8] have extensively modified CLIP's visual encoder without requiring retraining,

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The authors are with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: madandan@nwpu.edu.cn).

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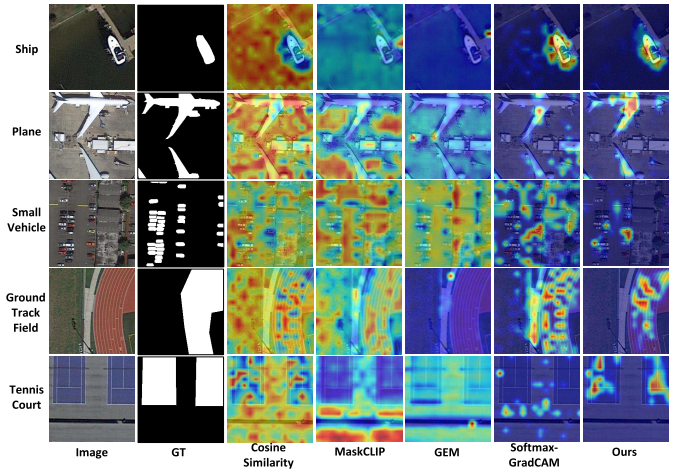



Fig. 1. Vision–language prior visualizations. The effective information from the cosine similarity prior remains implicit. The localization capability of Softmax-GradCAM [6] exhibits a decline in remote sensing applications. MaskCLIP [7] and GEM [8] demonstrate near-complete failure in handling remote sensing images. Our method decouples implicit feature similarities, filtering out background interference to achieve precise target localization.

enabling it to extract dense semantic visual features. However, the effectiveness of these methods on remote sensing images has not been verified, as depicted in Fig. 1.

Enhancing the quality of guidance from support information to the query image is key to improving the performance of FSS. Currently, most FSS methods undergo extensive training on the base classes before being applied to new categories. This approach poses challenges in improving guidance quality: simple nonparametric metrics are hard to sufficiently mine clues from support data, whereas overly complex parameterized architectures can introduce bias toward the base classes, leading to overfitting. Zhang et al. [20] proposed upsampling support prototypes and concatenating them with query features, using convolutional calculations instead of nonparametric measures such as cosine similarity. Xie et al. [21] introduced a scale-aware graph neural network for more precise localization of query objects. Hong et al. [22] applied 4-D convolutions to the 4-D correlation maps derived from pixel-level nonparametric measurements to extract segmentation predictions. While these methods enhance the segmentation guidance from support features to query features, they often involve overly complex parameter structures or network inputs.

In light of the aforementioned issues, this article proposes the implicit CLIP prior decoupling network (ICPD-Net). Well-trained remote sensing-specific VLMs, RemoteCLIP [23], exhibit strong generalization capabilities in remote sensing image-level tasks; however, the metric priors obtained through cross-modal features are implicit in aiding category perception. To address this, this article designs class-agnostic foreground–background text prompts, utilizing an implicit prior decoupling module (IPDM) to obtain vision–language priors that possess generalized perception abilities toward target classes from the complex and confounded foreground–background vision–language similarities. Subsequently, to further alleviate the issue of intraclass variance, this article introduces a reliable feature fusion module (RFFM). This module locates the high-confidence portions of visual features based on vision–language priors and performs a weighted fusion of located features with the foreground part of support visual features, thereby enhancing self-matching information. Finally, this article puts forward a dual visual priors module (DVPM) that combines the conventional cosine similarity prior with the carefully designed rank information prior based on Euclidean distance and the normalized discounted cumulative gain (NDCG) algorithm to fully explore the relationship between support and query features. The rank information prior innovatively incorporates ranking information of support-query feature distances in the metric process, while leveraging the NDCG algorithm to strengthen the weighting mechanism for rank information. Extensive experiments conducted on standard FSS benchmarks such as iSAID-$5^i$ show significant improvements with the proposed method, offering a novel and effective solution for empowering VLMs in remote sensing FSS.

In summary, the main contributions are as follows.

1) This article presents the ICPD-Net, which introduces the concept of IPDM to disentangle the intricate and confounded similarities between foreground and background in vision–language tasks produced by VLMs. This approach aims to derive generalized vision–language priors, thereby mitigating intraclass variation in FSS within the context of remote sensing.
2) To effectively utilize vision–language priors for the reduction of intraclass variance, the present study introduces the RFFM. This module integrates high-confidence query features with supporting features, thereby augmenting the self-matching information within the metric space.
3) This work introduces the DVPM, which not only employs the traditional cosine similarity prior but also incorporates a ranking information prior derived from Euclidean distance and the NDCG algorithm. The integration of this additional prior enhances the interrelationships among visual features and elevates the quality of visual feature guidance.
4) Extensive experiments were conducted on two validated standard FSS benchmarks, iSAID-$5^i$ and DLRSD-$5^i$, confirming the effectiveness of the ICPD-Net and achieving the SOTA performance with significant improvement.

## II. Related Works

### A. Few-Shot Learning

Few-shot learning (FSL) aims to construct a classification model that can recognize new concepts and categories using a small number of samples, which has significant applications in scenarios with limited sample sizes. Many methods in FSL provide insights for FSS tasks. FSL can be divided into two categories [17]: visual-based methods and semantic-based methods. The main idea of visual-based methods is to extract class-related cues from images for classification, which can be technically divided into optimization-based and metric-based approaches. Optimization-based methods [24] aim to rapidly fine-tune part of the network parameters using a small number of samples. Metric-based methods [25] design metric

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constraints to ensure smaller distances or higher similarities between features of the same class. Due to the limited sample size, the former is prone to overfitting, while the latter is heavily affected by intraclass variations. Semantic-based methods [26] aim to mine information from the text modality to supplement visual cues, thereby improving the performance of FSL. Chen et al. [26] adaptively mixed visual prototypes and semantic prototypes, demonstrating that semantic features from text are a powerful information source for FSL. Xu and Le [27] utilized text embeddings and conditional variational autoencoders to generate representative visual samples. Recently, large-scale pretrained VLMs such as CLIP [18] have demonstrated their strong generalization performance in FSL. Zhou et al. [28] improved the performance of CLIP in FSL through prompt learning. Gao et al. [29] proposed CLIP-Adapter to quickly adjust the outputs of the visual and text parts. Zhang et al. [17] demonstrated that a straightforward alignment of visual and textual features can effectively alleviate intraclass variation issues in FSL without necessitating additional fine-tuning of VLMs.

### B. Few-Shot Semantic Segmentation

FSS aims to achieve segmentation of novel categories using a small number of annotated images. The primary concept of current FSS approaches involves mining visual cues from support images to guide the segmentation of query images. Existing methods can be categorized into prototype-based and correlation-based approaches based on their guidance mechanisms. Prototype-based methods [11] construct representative features (i.e., prototypes) for specific classes from support images, then perform segmentation by measuring similarity with query features through parameter-free or parametric approaches. Prototypes can be computed using operations like clustering or average pooling, which are simple and robust but exhibit limited fine-grained perception. To address this, multiprototypes [30], holistic prototypes [13], and learnable prototypes [31] have been proposed. Liang et al. [32] utilized self-attention and cross-attention modules, respectively, employed to extract information about unseen categories distinct from background categories and to generate high-quality query attention maps. Correlation-based methods [33] preserve more fine-grained information by computing pixel-level dense relationships between query and support features. Mechanisms such as 4-D convolution [34], graph attention mechanism [35], and cycle-consistent transformer [36] have been developed for correlation analysis. Recently, with the advancement of pretrained models, researchers have attempted to integrate knowledge from large-scale pretrained models into FSS. For instance, Li et al. [37] introduced the masks generated by the segment anything model (SAM) [38] as prior information in remote sensing FSS, leveraging intramask consistency to alleviate intraclass variability. Additionally, some methods have incorporated VLMs into FSS tasks. Chen et al. [19] employed cosine similarity between visual features and language embeddings as class-agnostic priors, assembling different priors through a complex transformer-based network. Wang et al. [39] utilized the Softmax-GradCAM [6] of the CLIP [18] model classification scores as a localization prior. However, the effectiveness of these methods in remote sensing FSS remains unverified. The HSE [40] method represents the first attempt to introduce textual modality information in remote sensing FSS tasks. Nevertheless, its approach of embedding class-specific semantic knowledge into visual features at the feature level risks overfitting on base classes.

The proposed method, ICPD-Net, falls under the category of correlation-based approaches. It introduces RemoteCLIP [23] into FSS to mitigate intraclass variations, taking into account the complexity of remote sensing images. It decouples the semantic similarity between foreground and background categories, resulting in class-agnostic vision–language priors.

### C. VLMs for Semantic Segmentation

VLMs like CLIP [18] possess powerful instance-level vision–language alignment capabilities, making them well-suited for downstream tasks such as zero-shot classification [41], few-shot classification [42], and image–text retrieval [43]. Although not natively applicable to dense prediction tasks, several works [44], [45] have made significant attempts to effectively adapt VLMs for semantic segmentation tasks and enhance their performance. In weakly supervised semantic segmentation, Lin et al. [6] introduced Softmax-GradCAM as supervision signals. Wang et al. [46] fine-tuned CLIP on remote sensing images to extract class activation maps and refined the results using the SAM [38]. Xie et al. [47] employed CLIP as a supervisor to measure contrastive loss for evaluating segmentation performance. In open-vocabulary semantic segmentation (OVSS), Liang et al. [48] proposed mask prompt tuning to better adapt VLMs to mask proposal classification, thereby improving OVSS performance. Ghiasi et al. [49] adopted a similar strategy by leveraging class-agnostic masks and image–text pair data to fine-tune VLMs. Luo et al. [50] proposed SegCLIP, which trains the CLIP model on image–text pairs by combining contrastive loss, reconstruction loss, and superpixel-based KL loss. Mukhoti et al. [51] fine-tuned the CLIP model using patch-aligned contrastive learning. Some OVSS methods adopt training-free approaches: Li et al. [52] proposed CLIP feature surgery to identify and remove redundant features through average feature computation along the class dimension, alleviating noisy activation issues. Bousselham et al. [8] developed a grounding everything module utilizing self-attention mechanisms to improve dense visual features. Zhou et al. [7] proposed MaskCLIP by removing the final MLP layer and using the final value projection for dense features. In FSS, several methods have been inspired by these approaches: Wang et al. [53] and Chang et al. [54] computed cosine similarity between MaskCLIP [7] features and text embeddings to extract cross-modal priors.

The above methods enhance the application of VLMs in fine-grained perception tasks from different perspectives. The IPDM we proposed provides a novel and effective foreground–background decoupling mechanism for remote sensing FSS.

## III. Methodology

### A. Task Description

FSS aims to segment new categories using a handful of annotated images. Typically, standard FSS requires a base

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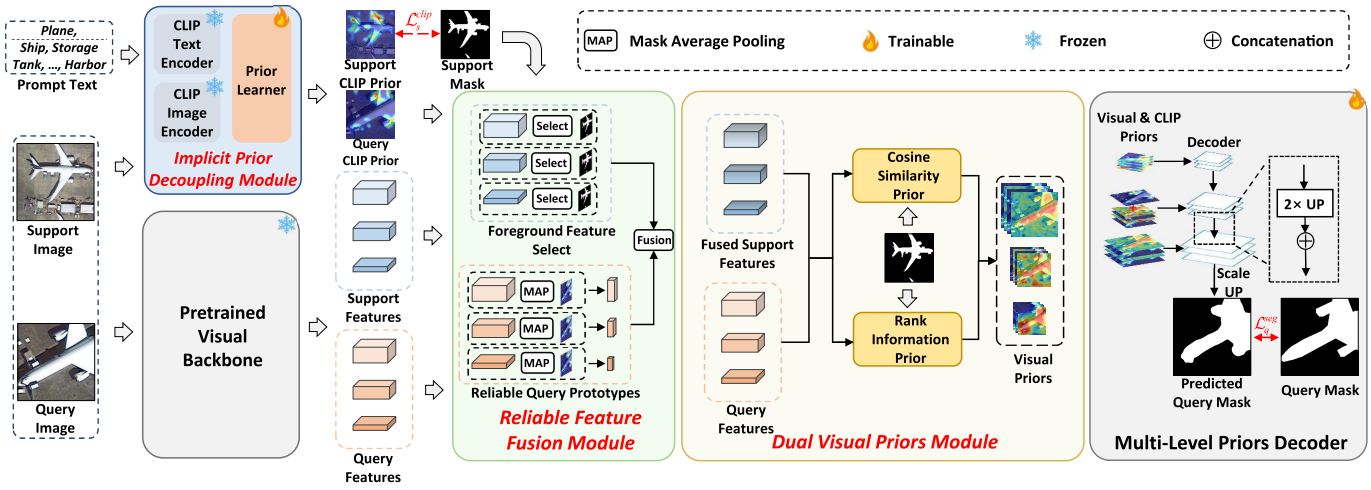



Fig. 2. Overall pipeline of our proposed ICPD-Net for the one-shot setting. Our innovative contributions include three core modules: the IPDM, RFFM, and DVPM. First, the support image, query image, and prompt text are fed into the IPDM (detailed in Fig. 3) to compute cross-modal CLIP priors. Subsequently, both the support and query images pass through a visual backbone network to extract multilevel support/query visual features. The RFFM then leverages CLIP priors to selectively blend high-confidence query features with support features. In the DVPM, multilevel features undergo visual metric computation to generate two types of priors: cosine similarity priors and rank information priors. Finally, a multilevel priors decoder integrates all generated priors to produce the predicted segmentation map.

dataset $\mathcal{D}_b$ that enables dense training, where images in the base dataset have high-quality annotations for base categories $\mathcal{C}_b$. A class-agnostic model $f_\theta$ is trained on the base dataset $\mathcal{D}_b$ to achieve good generalization on novel dataset $\mathcal{D}_n$ involving novel categories $\mathcal{C}_n$, with the constraint that base categories $\mathcal{C}_b$ and novel categories $\mathcal{C}_n$ are mutually disjoint, that is, $\mathcal{C}_b \cap \mathcal{C}_n = \emptyset$.

In FSS, the episodic learning paradigm of meta-learning is widely adopted, which partitions the dataset into episodic tasks. The model is trained on episodic tasks of base classes and tested on episodic tasks of novel classes. The episodic tasks in FSS follow the $K$-shot paradigm. Specifically, an episodic task consists of a support set $\mathcal{S} = \{(\boldsymbol{I}_s^i, \boldsymbol{M}_s^i)\}_{i=1}^K$ and a query set $\mathcal{Q} = \{(\boldsymbol{I}_q, \boldsymbol{M}_q)\}$. $\boldsymbol{I}_s$, $\boldsymbol{M}_s$, $\boldsymbol{I}_q$, and $\boldsymbol{M}_q$ represent the support image, support ground-truth mask, query image, and query ground-truth mask, respectively. Both $\mathcal{S}$ and $\mathcal{Q}$ encompass the same target class $\mathcal{C}_{\text{target}}$. The model $f_\theta$ is trained on the base dataset $\mathcal{D}_b$ with the following objective [19]:

$$\theta^* = \arg\min_\theta \mathcal{L}\left(\boldsymbol{M}_q, f_\theta\left(\boldsymbol{I}_s, \boldsymbol{M}_s, \boldsymbol{I}_q\right)\right) \tag{1}$$

where $\mathcal{L}$ is the training loss function and $\theta^*$ denotes the optimal parameters of model $f_\theta$. After training on the base dataset $\mathcal{D}_b$, inference is performed on the novel dataset $\mathcal{D}_n$ to obtain predicted masks, that is, $\tilde{\boldsymbol{M}}_q = f_{\theta^*}(\boldsymbol{I}_s, \boldsymbol{M}_s, \boldsymbol{I}_q)$.

### B. Methodology Overview

In this work, we introduce a novel approach that effectively incorporates VLMs into FSS for remote sensing, named the ICPD-Net. The ICPD-Net addresses two key challenges in FSS: 1) intraclass variations caused by limited labeled samples: the ICPD-Net leverages the generalization capability of VLMs through the IPDM and RFFM, overcoming the inherent limitations of VLMs in dense prediction tasks; and 2) low-quality guidance from visual features: the ICPD-Net proposes the DVPM, which combines the cosine similarity prior and the carefully designed rank information prior to thoroughly exploring the metric relationships between query and support features. As illustrated in Fig. 2, the core innovations of the ICPD-Net lie in the IPDM, RFFM, and DVPM.

Specifically, for an episodic task concerning category $\mathcal{C}_{\text{target}}$, the support image $\boldsymbol{I}_s$ and the query image $\boldsymbol{I}_q$ are first utilized to compute their vision–language priors through IPDM, that is, $\boldsymbol{Prior}_s^{\text{clip}}$ and $\boldsymbol{Prior}_q^{\text{clip}}$. And then, the support image $\boldsymbol{I}_s$ and query image $\boldsymbol{I}_q$ undergo a frozen pretrained visual backbone to obtain multilevel visual features $\boldsymbol{F}_s$ and $\boldsymbol{F}_q$. Using RFFM, the high-confidence portion of the query feature's vision–language prior $\boldsymbol{Prior}_q^{\text{clip}}$ is fused with the support target features. The fused support features $\hat{\boldsymbol{F}}_s$ and query features $\boldsymbol{F}_q$ are then input into DVPM to calculate dual visual priors, $\boldsymbol{Prior}_q^{\cos}$ and $\boldsymbol{Prior}_q^{\text{rank}}$. Finally, a multilevel priors decoder is used to obtain the predicted segmentation result $\tilde{\boldsymbol{M}}_q$ for the query image. In the following sections, we will sequentially introduce the IPDM (Section III-C), RFFM (Section III-D), and DVPM (Section III-E) of the ICPD-Net, as well as the multilevel priors decoder (Section III-F). For simplicity, the method will be explained in the context of a one-shot setup in the following.

### C. Implicit Prior Decoupling Module

Since VLMs like CLIP [18] do not natively possess fine-grained positional awareness, the proposed IPDM addresses this by decoupling the ambiguous vision–language similarities between foreground (target) and background (nontarget) categories. This process yields vision–language priors, where the high-confidence portions of these priors represent areas in the image more likely to be the target class. As a supplement to visual metrics methods, this prior is crucial for mitigating intraclass differences.

First, let us take a brief review of CLIP. The CLIP model consists of a visual encoder and a language encoder. Typically, the visual encoder and the language encoder are a Vision Transformer (ViT) [55] and a language Transformer based [56] on the attention mechanism, respectively. When an image

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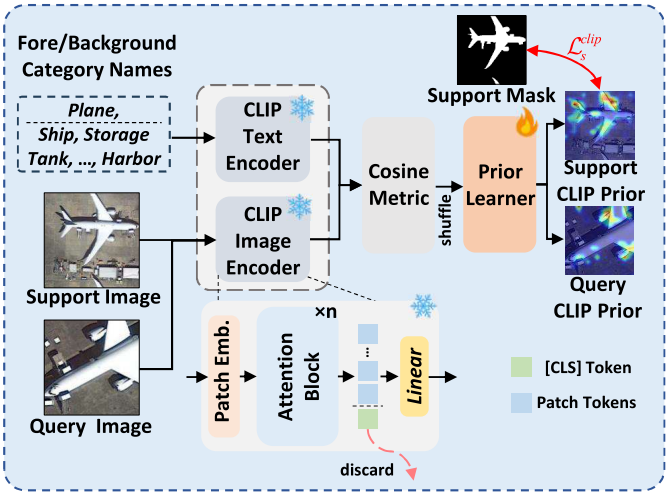



Fig. 3. Detailed description of the IPDM. This module employs the frozen CLIP image encoder and text encoder to encode the support/query images and foreground–background text prompts, respectively. Subsequently, their similarities are computed, based on which class-agnostic prompts are constructed. The trainable prior learner is then utilized to mine implicit CLIP priors.

and text that match each other are inputted, visual embedding $\boldsymbol{E}_v \in \mathbb{R}^{1\times C^{\text{clip}}}$ and text embedding $\boldsymbol{E}_t \in \mathbb{R}^{1\times C^{\text{clip}}}$ are obtained, which are well aligned in metric space. Specifically, for CLIP's visual encoder, the brief process of the model is shown in the following formula:

$$\begin{aligned}\boldsymbol{X}^0 &= \left[\boldsymbol{G}^0, \boldsymbol{T}_1^0, \boldsymbol{T}_2^0, \ldots, \boldsymbol{T}_{H\times W}^0\right]\\ \boldsymbol{X}^l &= \text{Layer}^l\left(\boldsymbol{X}^{l-1}\right)\\ &= \left[\boldsymbol{G}^l, \boldsymbol{T}_1^l, \boldsymbol{T}_2^l, \ldots, \boldsymbol{T}_{H\times W}^l\right]\\ \boldsymbol{E}_v &= \text{LinearProj}\left(\boldsymbol{G}^{\text{final}}\right).\end{aligned} \tag{2}$$

In this context, let $\boldsymbol{G}^{l-1} \in \mathbb{R}^{1\times C^{\text{emb}}}$ denote the [CLS] token, while $\boldsymbol{T}_i^{l-1} \in \mathbb{R}^{1\times C^{\text{emb}}}$ represents the patch tokens after the embedding of the image. Together, these elements form the input sequence $\boldsymbol{X}^{l-1}$ for the $l$th layer of the CLIP ViT encoder. The term Layer$^l$ refers to the multihead attention mechanism implemented at the $l$th layer. As illustrated in Fig. 3, to acquire spatially aware information, we omit $\boldsymbol{G}^{\text{final}}$ and instead apply a projection to $\boldsymbol{T}_i^{\text{last}}$ using the LinearProj function, despite the uncertain efficacy of this projection. Following the concatenation and reshaping of the projected tokens, we derive the CLIP visual feature map $\boldsymbol{F}_v \in \mathbb{R}^{H\times W\times C^{\text{clip}}}$. The vision–language similarity map $\boldsymbol{S} \in [0,1]^{H\times W\times 1}$ is computed based on the cosine similarity metric as follows:

$$\boldsymbol{S} = \phi^{-1}\left[\text{COS}\left(\phi\left(\boldsymbol{F}_v\right), \boldsymbol{E}_t\right)\right] \tag{3}$$

where $\phi$ and $\phi^{-1}$ denote transformation operation $\mathbb{R}^{H\times W\times C} \to \mathbb{R}^{HW\times C}$ and its inverse operation $\mathbb{R}^{HW\times C} \to \mathbb{R}^{H\times W\times C}$, respectively. In remote sensing images, there are many background classes (nontarget classes) beyond the foreground class (target class). Vision–language information about the background classes can provide cues for the segmentation of the foreground. To this end, we construct vision–language prompt $\boldsymbol{P} \in [0,1]^{H\times W\times(1+N_{\text{back}})}$ for the foreground and background as follows:

$$\boldsymbol{P} = \text{CONCAT}\left[\boldsymbol{S}_{\text{fore}}, \boldsymbol{S}_{\text{back}}^1, \ldots, \boldsymbol{S}_{\text{back}}^{N_{\text{back}}}\right] \tag{4}$$

where $\boldsymbol{S}_{\text{fore}}$ and $\boldsymbol{S}_{\text{back}}^i$ are the text embeddings for the target class and background classes, respectively, obtained through (3). CONCAT[·] is a tensor concatenation operation that concatenates $\boldsymbol{S}_{\text{fore}}$ and $\boldsymbol{S}_{\text{back}}$ along the channel dimension. Under the meta-learning paradigm, fixed background class ordering introduces distribution discrepancies in similarity data between base and novel classes, thereby inducing class correlations that further exacerbate overfitting. Shuffling the background class similarity scores effectively removes the influence of class-specific information. To eliminate class-specific information and make the IPDM as class-agnostic as possible, we shuffle the order of background similarity

$$\boldsymbol{P}[:,:,1:] = \text{shuffle}\left(\boldsymbol{P}[:,:,1:]\right). \tag{5}$$

To mine vision–language prior focused on the target class from $\boldsymbol{P}$, this article designs a prior learner for prior decoupling. The vision–language prompt $\boldsymbol{P}$, after being processed through this network, yields the vision–language prior $\boldsymbol{Prior}^{\text{clip}} \in [0,1]^{H\times W\times 1}$ for the target class. The process is as follows:

$$\begin{aligned}\boldsymbol{X} &= \text{ReLU}\left(\mathcal{N}\left(\text{convT}_1\left(\boldsymbol{P}\right)\right)\right)\\ \boldsymbol{X} &= \text{ReLU}\left(\mathcal{N}\left(\text{convT}_2\left(\boldsymbol{X}\right)\right)\right)\\ \boldsymbol{X} &= \text{MaxPool}\left(\text{ReLU}\left(\mathcal{N}\left(\text{conv}_1\left(\boldsymbol{X}\right)\right)\right)\right)\\ \boldsymbol{X} &= \text{MaxPool}\left(\text{ReLU}\left(\mathcal{N}\left(\text{conv}_2\left(\boldsymbol{X}\right)\right)\right)\right)\\ \boldsymbol{Prior}^{\text{clip}} &= \zeta\left(\text{conv}_3\left(\boldsymbol{X}\right)\right)\end{aligned} \tag{6}$$

where convT, conv, ReLU, and MaxPool represent 2-D transposed convolution, 2-D convolution, rectified linear unit, and max-pooling operation, respectively. $\mathcal{N}$ stands for instance normalization, and $\zeta$ is the Sigmoid activation function, ensuring that the output values are within the range of [0, 1]. The reason for adopting instance normalization in the prior learner is that it performs normalization along the channel dimension for individual samples, preserving the independence of different classes and samples. To ensure that the parts of $\boldsymbol{Prior}^{\text{clip}}$ with higher values reflect a higher probability of being the target class, and otherwise are more likely to be background classes, we use the support image annotation $\boldsymbol{M}_s \in \{0,1\}^{H\times W\times 1}$ as a constraint and calculate the prior loss as follows:

$$\mathcal{L}_s^{\text{clip}} = \text{BCE}\left(\boldsymbol{Prior}_s^{\text{clip}}, \boldsymbol{M}_s\right) \tag{7}$$

where BCE is the binary cross-entropy loss.

### D. Reliable Feature Fusion Module

Using the information from the query image itself to guide its segmentation can help improve the model's robustness and alleviate intraclass variations [16]. The conventional approach [15] is to use the parts of the support-query metric results that are above a certain threshold for self-support. However, if the support query discrepancy is too large, this method is likely to fail. To address this, our proposed RFFM utilizes the vision–language priors of the query image instead of the support-query metric results. This not only further encourages the application of VLMs in FSS but also further mitigates the issue of intraclass variations. Specifically, the vision–language

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prior $\boldsymbol{Prior}_q^{\text{clip}}$ of the query image $\boldsymbol{I}_q$ are first normalized using a min-max normalization operation

$$\widehat{\boldsymbol{Prior}}_q^{\text{clip}} = \frac{\boldsymbol{Prior}_q^{\text{clip}} - \min\left(\boldsymbol{Prior}_q^{\text{clip}}\right)}{\max\left(\boldsymbol{Prior}_q^{\text{clip}}\right) - \min\left(\boldsymbol{Prior}_q^{\text{clip}}\right) + \epsilon} \tag{8}$$

where $\epsilon$ is a very small term to prevent division by zero. Based on the normalized prior $\widehat{\boldsymbol{Prior}}_q$, thresholding is performed: pixels above the threshold are considered high-confidence and set to 1; otherwise, they are set to 0

$$\widetilde{\boldsymbol{M}}_q = \mathbf{1}\left(\widehat{\boldsymbol{Prior}}_q^{\text{clip}} > \tau\right) \tag{9}$$

where $\tau$ is set to 0.7, representing the confidence threshold, while $\mathbf{1}$ denotes the indicator function.

The support visual feature map $\boldsymbol{F}_s \in \mathbb{R}^{H\times W\times C}$ and query visual feature map $\boldsymbol{F}_q \in \mathbb{R}^{H\times W\times C}$ at the same level will be processed through the RFFM. First, the query prototype $\mathcal{P}_q$ is computed based on $\widetilde{\boldsymbol{M}}_q$ and $\boldsymbol{F}_q$

$$\begin{aligned}\mathcal{P}_q &= \text{MAP}\left(\widetilde{\boldsymbol{M}}_q, \boldsymbol{F}_q\right) \\ &= \frac{\sum_{i,j} \widetilde{\boldsymbol{M}}_q(i,j)\cdot \boldsymbol{F}_q(i,j)}{\sum_{i,j} \widetilde{\boldsymbol{M}}_q(i,j)}\end{aligned} \tag{10}$$

where MAP denotes the mask average pooling operation. Based on the support image's ground truth annotation $\boldsymbol{M}_q$, locate the target class region in the support features. Perform a weighted fusion of the features from this region with the query prototype $\mathcal{P}_q$

$$\boldsymbol{F}_s(i,j) = (1-\alpha)\cdot \boldsymbol{F}_s(i,j) + \alpha \cdot \boldsymbol{P}_q \quad \text{if } \boldsymbol{M}_s(i,j) = 1 \tag{11}$$

where $\alpha$ is determined to be 0.5 according to experimental findings, representing the balancing weight in the fusion.

### E. Dual Visual Priors Module

In addition to the cross-modal prior, the guidance of support visual cues is equally crucial for FSS. Based on the dense pixel correlations between support and query images, we employ the cosine similarity prior $\boldsymbol{Prior}^{\cos}$, which has demonstrated effectiveness [10], along with a carefully designed rank information prior $\boldsymbol{Prior}^{\text{rank}}$. These constitute the DVPM.

For the cosine similarity prior $\boldsymbol{Prior}^{\cos}$, specifically, the support feature $\boldsymbol{F}_s$ is first masked using the ground-truth annotation $\boldsymbol{M}_s$

$$\hat{F}_s = F_s \odot M_s \tag{12}$$

where $\odot$ represents the Hadamard product, and then cosine similarity is calculated between $\hat{\boldsymbol{F}}_s$ and $\boldsymbol{F}_q$

$$(\boldsymbol{S}_{\cos})_{ij} = \text{ReLU}\left(\frac{\boldsymbol{F}_q(i)\cdot \hat{\boldsymbol{F}}_s(j)}{\left\|\boldsymbol{F}_q(i)\right\| \times \left\|\hat{\boldsymbol{F}}_s(j)\right\|}\right) \tag{13}$$

where $\boldsymbol{S}_{\cos} \in [0,1]^{HW\times HW}$. Then, perform a masked average pooling on $\boldsymbol{S}_{\cos}$ along the support dimension and conduct a dimension transformation to obtain the cosine similarity prior $\boldsymbol{Prior}^{\cos} \in [0,1]^{H\times W\times 1}$

$$\boldsymbol{Prior}^{\cos} = \phi^{-1}\left(\frac{\sum_{i=1}^{HW} \boldsymbol{S}_{\cos}(:,i)}{\sum_{j=1}^{HW} \boldsymbol{M}_s(j)}\right). \tag{14}$$

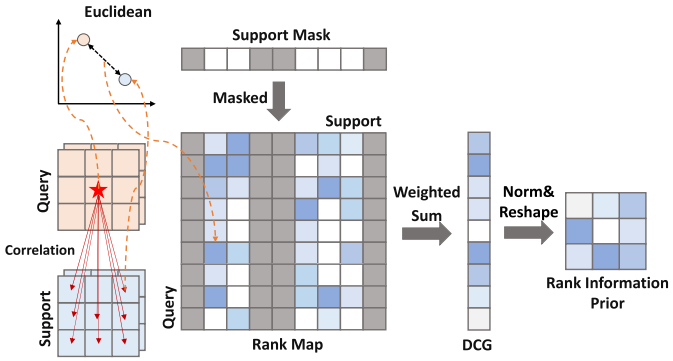



Fig. 4. Detailed description of the rank information prior. First, the Euclidean distance relationships between support features and query features are computed. Based on these distance relationships, ranking weights are considered to calculate the DCG for each query feature. The NDCG is then derived through normalization, thereby obtaining the rank information prior.

While the cosine metric is simple and effective, solely using the cosine similarity prior $\boldsymbol{Prior}^{\cos}$ fails to fully exploit the support-query visual cues. To address this, we design the rank information prior $\boldsymbol{Prior}^{\text{rank}}$ by leveraging the NDCG [57] algorithm on ranking relationships derived from Euclidean distances between support and query features. This mechanism prioritizes guidance from support-query pairs with geometrically meaningful similarities, thereby enhancing visual grounding through structured rank.

NDCG is a metric used to evaluate the effectiveness of ranking, which is calculated based on the relevance of the ranked results and takes into account the position of the results in the ranking list. The higher the value, the more reliable the ranking effect is indicated to be. In FSS, the distance relationships and their order among support query features contain crucial information. As Fig. 4, for a query pixel-level feature, its distances to support pixel-level features are ranked. If the closer query pixel-level features predominantly belong to the target class, it suggests that the query feature is more likely to belong to that class. Based on this idea, we designed the rank information prior. First, calculate the pixel-wise Euclidean distance $\boldsymbol{D} \in \mathbb{R}_+^{HW\times HW}$ between the support features and the query features

$$(\boldsymbol{D})_{ij} = \left\|\boldsymbol{F}_q(i) - \boldsymbol{F}_s(j)\right\|_2. \tag{15}$$

Then, calculate the number of foreground pixels $k = \sum \boldsymbol{M}_s$ in $\boldsymbol{M}_s$. For $\boldsymbol{F}_q(i)$, find the coordinates of the $k$ support features that are closest in distance to it

$$\text{indices} = \text{topk}\left(-\boldsymbol{D}(i,:)\right) \tag{16}$$

where topk$(\cdot)$ is to find the indices of the largest $k$ elements. Then, based on whether they fall within the foreground, correlation scores vector $\boldsymbol{rel} = \boldsymbol{M}_s(\text{indices})$ are obtained: if they fall within the foreground, the correlation score is 1; otherwise, it is 0. Here, $\boldsymbol{rel} \in \{0,1\}^{1\times k}$. Considering the impact of the ranking position, calculate the discounted cumulative gain (DCG)

$$\text{DCG} = \sum_{i=1}^{k} \frac{2^{\boldsymbol{rel}(i)} - 1}{\log_2(i+1)}. \tag{17}$$

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The ideal DCG IDCG for the target class is

$$\mathrm{IDCG} = \sum_{i=1}^{k} \frac{1}{\log_2(i+1)} \tag{18}$$

which represents that the closest $k$ matched features all fall within the target class positions. Thus, the NDCG is calculated as follows:

$$\mathrm{NDCG} = \frac{\mathrm{DCG}}{\mathrm{IDCG}}. \tag{19}$$

The closer the NDCG is to 1, the more likely the feature $\boldsymbol{F}_q(i)$ is to belong to the target class. The rank information prior $\boldsymbol{Prior}^{\mathrm{rank}} \in [0,1]^{H\times W\times 1}$ for $\boldsymbol{F}_q$ is obtained based on the aforementioned algorithm.

### F. Multilevel Priors Decoder

The multilevel visual feature maps from the visual backbone, when processed through DVPM, yield multilevel visual priors, including cross-modal vision–language priors, forming a multilevel multimodal prior set $\mathcal{G} = \{\boldsymbol{Prior}_q^{\mathrm{clip}}, \boldsymbol{Prior}_1^{\mathrm{cos}}, \ldots, \boldsymbol{Prior}_l^{\mathrm{ndcg}}\}$. To integrate the multiscale prior information for the final segmentation result, we constructed a commonly used multilevel priors decoder as shown in Fig. 2. This decoder primarily decodes the priors through convolution and integrates them using upsampling and cross-layer connections to produce the final prediction $\hat{\boldsymbol{Y}} \in [0,1]^{H_{\mathrm{img}}\times W_{\mathrm{img}}\times 2}$, with two channels representing the probabilities of target and nontarget classes, respectively. To enhance the model's sensitivity to small targets, we optimize the prediction results using SACELoss proposed by Ao et al. [10], leading to the final training loss of

$$\begin{aligned} \mathcal{L}_q^{\mathrm{seg}} &= \mathrm{SACE}\left(\hat{\boldsymbol{M}}_q, \boldsymbol{M}_q\right) \\ \mathcal{L}_{\mathrm{total}} &= \mathcal{L}_s^{\mathrm{clip}} + \mathcal{L}_q^{\mathrm{seg}}. \end{aligned} \tag{20}$$

Extending the model to $K$-shot is straightforward. Simply average the prior sets $\mathcal{G}$ obtained from different support images, then feed the result into the decoder to predict the segmentation result

$$\mathcal{G} = \frac{1}{K}\sum_{i}^{K} \mathcal{G}_i. \tag{21}$$

Here, $\mathcal{G}_i$ is the prior set from the support image $\boldsymbol{I}_s^i$.

## IV. Experimental Settings

In this section, comprehensive experiments are conducted to validate the effectiveness of our proposed method. First, we introduce the datasets used in this study: iSAID-$5^i$ and DLRSD-$5^i$. Next, the evaluation metric and implementation details are described.

### A. Datasets

*1) iSAID-$5^i$:* The iSAID-$5^i$ is one of the most popular datasets for remote sensing image FSS. It is created by Yao et al. [58] based on the iSAID dataset [59] following the setup of PASCAL-$5^i$ [9]. Specifically, the 15 object categories in the iSAID dataset are divided into three groups, each containing five categories, as detailed in Table I. The iSAID-$5^i$ dataset comprises 18 076 training images and 63 363 validation images, all with a resolution of 256 × 256. During cross-validation, one fold's categories are treated as novel classes for testing, while the remaining folds serve as base classes for training.

TABLE I
Testing Classes for Each Fold of iSAID-$5^i$

| Dataset | Test classes |
|---|---|
| iSAID-$5^0$ | ship, storage tank, baseball diamond, tennis court, basketball court |
| iSAID-$5^1$ | ground track field, bridge, large vehicle, small vehicle, basketball court |
| iSAID-$5^2$ | swimming pool, round pool, roundabout, soccer ball field, plane, harbor |

TABLE II
Testing Classes for Each Fold of DLRSD-$5^i$

| Dataset | Test classes |
|---|---|
| DLRSD-$5^0$ | airplane, bare soil, buildings, cars, chaparral |
| DLRSD-$5^1$ | court, dock, field, grass, mobile home |
| DLRSD-$5^2$ | pavement, sand, sea, ship, tanks |

*2) DLRSD-$5^i$:* Similar to iSAID-$5^i$, DLRSD-$5^i$ was created by Wang et al. [33] using another publicly available dataset, DLRSD [60]. Wang et al. [33] selected 15 semantic categories from the 21 covered by DLRSD and divided them into three folders, as detailed in Table II. All images have a resolution of 256 × 256, comprising a total of 2100 images. Similarly, employing cross-validation, one folder is designated for testing novel classes, while the categories in the remaining two folders serve as the base classes for training.

### B. Evaluation Metric

Our work primarily employs the mean intersection over union (mIoU) as the evaluation metric for quantitative analysis. For each category, IoU is calculated as IoU = TP/(TP + FP + FN), where TP, FP, and FN represent the true-positive, false-positive, and false-negative pixels in the prediction results, respectively. The mIoU metric is obtained by averaging the IoU values across all categories in the test dataset

$$\mathrm{mIoU} = \frac{1}{C}\sum_{c=1}^{C} \mathrm{IoU}_c \tag{22}$$

where $C$ denotes the number of categories in the dataset. To further evaluate the effectiveness of each module, we employ foreground–background IoU (FB-IoU) as an additional evaluation metric in the ablation study to further validate the modules' contributions

$$\mathrm{FB\text{-}IoU} = \frac{1}{2}\left(\mathrm{IoU}_f + \mathrm{IoU}_b\right) \tag{23}$$

where $\mathrm{IoU}_f$ represents the IoU value for all foreground (target) classes and $\mathrm{IoU}_b$ corresponds to the IoU value for the background (nontarget) class.

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TABLE III

PERFORMANCE COMPARISON UNDER ONE-SHOT AND FIVE-SHOT SETTINGS ON THE iSAID-$5^i$ DATASET, MEASURED BY MIOU (%). BOLD RESULTS INDICATE THE BEST PERFORMANCE, AND UNDERLINED RESULTS DENOTE THE SECOND-BEST. METHODS MARKED WITH † INCORPORATE ADDITIONAL MODEL INFORMATION BEYOND THE BACKBONE, SIMILAR TO OUR APPROACH

| Methods | 1-shot | | | | 5-shot | | | |
|---|---|---|---|---|---|---|---|---|
| | Fold-0 | Fold-1 | Fold-2 | Mean | Fold-0 | Fold-1 | Fold-2 | Mean |
| VGG16 | | | | | | | | |
| CANet [20] | 19.73 | 17.98 | 30.93 | 22.88 | 23.45 | 20.53 | 30.12 | 24.70 |
| PFENet [12] | 28.57 | 23.38 | 27.77 | 26.57 | 28.02 | 27.23 | 28.14 | 27.80 |
| HSNet [22] | 22.74 | 23.05 | 25.76 | 23.84 | 27.20 | 28.86 | 28.82 | 28.29 |
| SCCNet [33] | 35.71 | 30.33 | 36.68 | 34.24 | 40.40 | 32.56 | 39.31 | 37.42 |
| PCFNet [10] | 39.93 | 27.42 | 33.55 | 33.63 | 50.73 | 34.48 | 44.25 | 43.12 |
| HRL [64] | 33.54 | 23.10 | 31.58 | 29.40 | 43.02 | 26.19 | 39.26 | 36.16 |
| MGCNet†[37] | 38.24 | 28.61 | 40.21 | 35.69 | 43.81 | 31.07 | 48.31 | 41.06 |
| HSENet† [40] | 37.18 | 29.79 | 36.04 | 34.34 | 40.87 | 30.20 | 43.44 | 38.17 |
| Ours | **56.27** | **35.55** | **50.20** | **47.34** | **61.21** | **42.86** | **56.42** | **53.50** |
| ResNet-50 | | | | | | | | |
| CANet [20] | 18.80 | 15.62 | 25.79 | 20.07 | 23.86 | 18.54 | 32.00 | 24.80 |
| PFENet [12] | 35.84 | 23.35 | 27.20 | 28.80 | 42.42 | 25.34 | 33.00 | 33.59 |
| HSNet [22] | 30.76 | 24.35 | 38.20 | 31.10 | 38.08 | 30.56 | 45.28 | 37.79 |
| SCCNet [33] | 40.74 | 31.25 | 46.40 | 39.45 | 44.27 | 31.62 | 50.45 | 42.11 |
| PCFNet [10] | 42.94 | 27.17 | 40.06 | 36.72 | 52.97 | 34.91 | 51.62 | 46.50 |
| $MS^2A^2$Net[65] | 34.96 | 25.30 | 30.71 | 30.32 | 42.30 | 27.10 | 36.31 | 35.24 |
| SSMP [15] | 44.08 | 32.21 | 44.11 | 39.82 | 57.95 | 39.63 | 53.99 | 50.52 |
| HRL [64] | 41.50 | 27.84 | 34.51 | 34.62 | 49.01 | 32.86 | 43.26 | 41.71 |
| MGCNet†[37] | 42.77 | 30.59 | 46.39 | 39.92 | 49.14 | 32.58 | 52.75 | 44.82 |
| HSENet† [40] | 44.03 | 35.87 | 39.26 | 39.72 | 47.03 | 36.75 | 45.66 | 43.15 |
| Ours | **58.63** | **38.76** | **56.82** | **51.40** | **60.09** | **43.46** | **58.15** | **53.90** |

### C. Implementation Details

We utilize the RemoteCLIP [23] model, a CLIP [18] model pretrained on remote sensing data, with the visual encoder architecture being the ViT-L-14 variant. The visual encoder comprises 24 Transformer [56] layers, each containing 16 attention heads and an embedding dimension of 1024, yielding a final output dimension of 768. Images are first resized to $224 \times 224 \times 3$ pixels, ultimately generating CLIP visual feature maps of size $16 \times 16 \times 768$. Class names are directly fed into the text encoder to produce CLIP text features of size $1 \times 768$, where target classes are treated as foreground categories and all remaining classes as background categories. To capture multiscale hierarchical information, we conduct experiments using ResNet-50 [61] and VGG16 [62] pretrained on the ImageNet [63] as the backbones. To prevent learning class-specific features that could lead to overfitting on base classes, we freeze all parameters in both the VLMs and the visual backbones. Our method is implemented based on PyTorch. In our experiments, all images are fixed with $256 \times 256 \times 3$, with random horizontal and vertical flipping with a certain probability applied during training for data augmentation. We employ the Adam optimizer with a learning rate of $1e^{-4}$. For both the iSAID-$5^i$ and DLRSD-$5^i$ datasets, we train the model for 100 epochs with a batch size of 8. All experiments are conducted on a single NVIDIA RTX 4090 GPU.

## V. RESULTS AND DISCUSSIONS

We evaluate the experimental results by comparing them with state-of-the-art methods. Thorough ablation studies are performed to analyze the role of each component of the proposed method. Finally, we discuss the limitations and present the prospects for future work.

### A. Comparison With SOTA

To validate the effectiveness of the proposed method, we report its performance under various settings on the iSAID-$5^i$ and DLRSD-$5^i$ datasets, with comparisons against SOTA methods. Note that our method incorporates external prior information from RemoteCLIP [23], while comparable approaches MGCNet [37] and HSENet [40] (annotated in Tables III and IV) similarly integrate external model knowledge: the MGCNet leverages mask priors from SAM [38], and the HSENet utilizes semantic embeddings from CLIP [18]. The compared methods include the CANet [20] (CVPR2019), PFENet [12] (T-PAMI2022), HSNet [22] (ECCV2022), SCCNet [33] (SIGSPATIAL2023), PCFNet [10] (T-GRS2023), $MS^2A^2$Net [65] (T-GRS2024), MGCNet [37] (T-GRS2024), HSENet [40] (T-GRS2025), SSMP [15] (T-GRS2024), and HRL [64] (T-GRS2025), where MGCNet [37] and HSENet [40], like our method, incorporate additional model information.

*1) Comparative Analysis on iSAID-$5^i$:* Our experimental results on iSAID-$5^i$ and those of comparative methods are presented in Table III. We adopt mIoU as the evaluation metric and conduct experiments under both one-shot and five-shot settings. The results of our method are shown in the "ours" row. As shown in the Table III, our method achieves comprehensive performance superiority across different settings, including the MGCNet [37] and HSENet [40]. Specifically, when using VGG16 [62] as the backbone, our method outperforms the second-best method MGCNet [37] by 11.65% in average mIoU across three folds under the one-shot setting and surpasses the second-best method PCFNet [10] by 10.38% under the five-shot setting. With ResNet-50 [61] as the backbone, our method leads the second-best method MGCNet

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TABLE IV

PERFORMANCE COMPARISON UNDER ONE-SHOT AND FIVE-SHOT SETTINGS ON THE DLRSD-$5^i$ DATASET, MEASURED BY MIOU (%), RESNET-50 IS USED AS THE BACKBONE. BOLD RESULTS INDICATE THE BEST PERFORMANCE, AND UNDERLINED RESULTS DENOTE THE SECOND-BEST. METHODS MARKED WITH † INCORPORATE ADDITIONAL MODEL INFORMATION BEYOND THE BACKBONE, SIMILAR TO OUR APPROACH

| Methods | 1-shot | | | | 5-shot | | | |
|---|---|---|---|---|---|---|---|---|
| | Fold-0 | Fold-1 | Fold-2 | Mean | Fold-0 | Fold-1 | Fold-2 | Mean |
| PFENet [12] | 21.98 | 20.67 | 11.17 | 17.94 | 24.12 | 24.76 | 16.01 | 21.63 |
| HSNet [22] | 22.00 | 47.20 | 34.73 | 34.64 | 27.46 | 52.32 | 46.23 | 42.00 |
| SCCNet [33] | 26.48 | 49.59 | 41.89 | 39.32 | 30.26 | 51.08 | 47.60 | 42.92 |
| PCFNet [10] | 31.09 | 37.52 | 41.89 | 36.83 | 36.67 | 42.70 | **54.20** | 44.52 |
| HRL [64] | **39.72** | 51.11 | 39.38 | 43.30 | **41.37** | 51.90 | 42.86 | 45.38 |
| MGCNet†[37] | 30.37 | 50.57 | 43.01 | 41.32 | 33.64 | 54.84 | 52.34 | 46.94 |
| Ours | 37.44 | **53.61** | **48.42** | **46.49** | 39.09 | **57.17** | 53.46 | **49.91** |

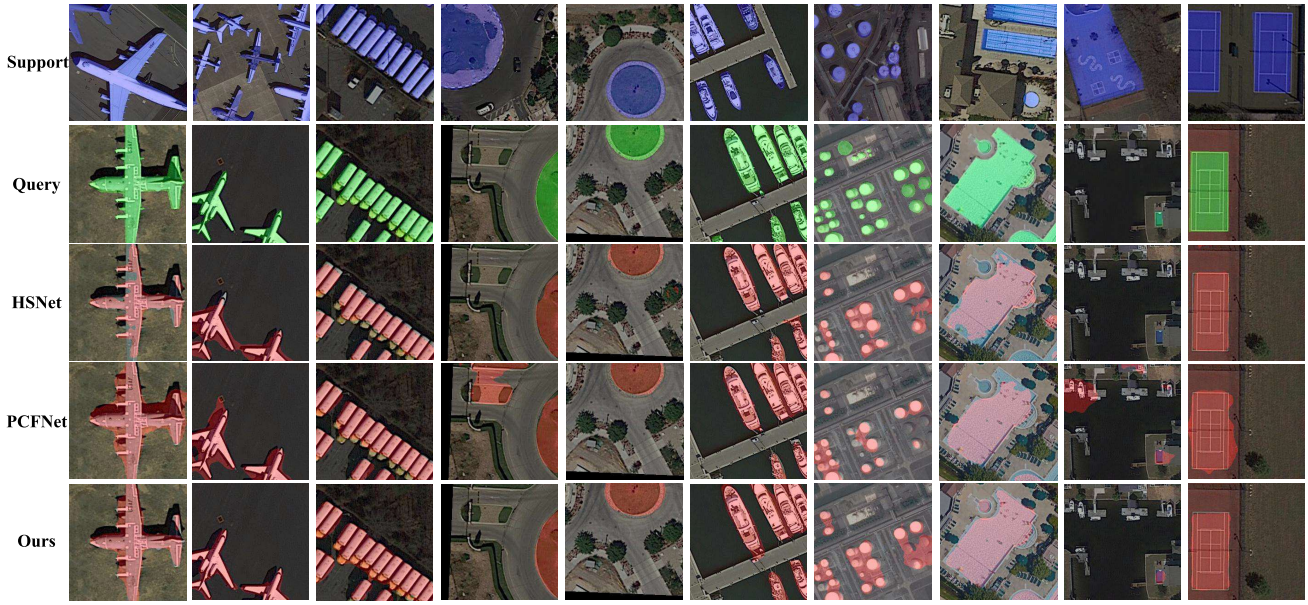



Fig. 5. Qualitative results of one-shot segmentation on the iSAID-$5^i$ dataset. (Top to bottom) Support image with GT (blue), query image with GT (green), the HSNet [22] prediction results (red), the PCFNet [10] prediction results (red), and our method's prediction results (red).

TABLE V

IMPACT ANALYSIS OF MODEL COMPONENTS. THE RESULTS ARE EVALUATED ON ISAID-$5^i$ USING MIOU (%) AND FB-IOU (%). GREEN RESULTS INDICATE THE EXTENT OF PERFORMANCE DEGRADATION AFTER REMOVING EACH MODULE. THE ABLATED MODULES INCLUDE IPDM, RFFM, COS, AND RANK, WHERE COS AND RANK REPRESENT THE PRIORS CONSTITUTING THE DVPM

| Modules | | | | 1-shot | | | | | 5-shot | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| IPDM | RFFM | COS | RANK | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU |
| ✓ | ✓ | ✓ | ✓ | **58.63** | **38.76** | **56.82** | **51.40** | **66.56** | 60.09 | **43.46** | **58.15** | **53.90** | **68.73** |
| ✓ | | ✓ | ✓ | 58.51 | 37.71 | 56.23 | 50.82 **-0.58** | 65.49 **-1.07** | **62.21** | 39.60 | 54.17 | 51.99 **-1.91** | 65.82 **-2.91** |
| | | ✓ | ✓ | 43.06 | 28.19 | 40.53 | 37.26 **-13.56** | 58.19 **-7.30** | 52.66 | 36.44 | 53.86 | 47.65 **-4.34** | 64.41 **-1.41** |
| | | ✓ | | 39.62 | 25.83 | 38.62 | 34.69 **-2.59** | 57.39 **-0.80** | 46.37 | 32.09 | 47.93 | 42.13 **-5.52** | 61.05 **-3.36** |

[37] by 11.48% under the one-shot setting. In the five-shot setting, our method exceeds the second-best method HRL [64] by 3.38%.

The significant performance improvement is primarily attributed to our proposed IPDM, particularly under the one-shot setting, as demonstrated in Table V. IPDM leverages extracted cross-modal priors to alleviate the substantial intra-class variations inherent in FSS. From the visualization results in Fig. 5, it can be observed that our method generates segmentation results with higher precision, particularly excelling in segmenting objects with large intraclass variations. For instance, in the second-to-last column depicting the category of "swimming pool," where significant scale variations are present, only our method accurately locates and segments the target. RFFM and DVPM also enhanced model performance, further alleviating intraclass variations and improving visual metric richness, resulting in better segmentation edges and suppressed erroneous predictions.

*2) Comparative Analysis on DLRSD-$5^i$:* Our experimental results on DLRSD-$5^i$ and those of comparative methods are presented in Table IV. On DLRSD-$5^i$, all methods adopt ResNet as the backbone. In Table IV, all approaches utilize

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ResNet-50 as the backbone, with mIoU as the evaluation metric under both one-shot and five-shot settings. Our method achieves the highest average mIoU under both configurations. Specifically, it outperforms the second-best method HRL [64] by 3.19% under the one-shot setting and outperforms the MGCNet [37] by 2.97% under the five-shot setting. The relatively moderate performance improvement on DLRSD-$5^i$ could be attributed to the limited base class samples in this dataset, which likely hinders effective parameter learning for IPDM's prior learner. Nevertheless, our method still outperforms traditional methods and the MGCNet [37] with SAM [38] incorporated, validating the effectiveness of our approach.

*3) Analysis of Inconsistent Improvements:* In our approach, the relative improvement is more significant in the one-shot setting. Specifically, on the iSAID-$5^i$ dataset in the one-shot scenario using ResNet50 as the backbone, our method achieves an improvement of over 10% compared to the suboptimal method. However, in the five-shot scenario, the improvement is relatively smaller at only 3.48%. We believe this occurs because, as the number of support images increases, intraclass variations become better addressed, contributing to cross-modal priors being less pronounced. Moreover, the degree of improvement achieved by our method varies across different folds within the same dataset. For example, under the one-shot setting, fold-0 achieves an mIoU improvement exceeding 14%, while fold-1 shows less than a 3% improvement. We attribute this to varying levels of intraclass variation and scale differences among classes across different folds. A similar phenomenon has also been observed in FSS involving natural images [39].

## B. Ablation Studies

We conduct extensive ablation experiments using ResNet-50 [61] as the backbone on the iSAID-$5^i$ dataset, along with testing different module configurations, to thoroughly analyze the effectiveness and contribution of each component.

*1) Effects of the Each Proposed Module:* We first conduct a detailed analysis of our proposed modules, with experimental results shown in Table V. Our framework primarily consists of IPDM, RFFM, and DVPM. Notably, the DVPM contains both a conventional cosine similarity prior and our carefully designed rank information prior. To specifically analyze our innovative contributions, we treat these two elements as separate components (abbreviated as COS and RANK) for ablation studies. Considering the interdependencies among components, we adopt a progressive removal approach starting from the complete method. The performance degradation at each removal stage quantitatively demonstrates each component's individual contribution.

As evidenced in Table V, under the one-shot setting, our proposed IPDM contributes most significantly to model performance. The third experimental row demonstrates that removing IPDM causes substantial metric degradation: 13.56% average mIoU and 7.30% FB-IoU reductions across three folds, with comparable declines in other fold-specific metrics. This indicates that IPDM substantially mitigates intraclass variation through cross-modal prior extraction, thereby enhancing FSS performance when support samples are extremely limited. In contrast, under the five-shot configuration, where increased support samples inherently alleviate intraclass variation, IPDM's contribution diminishes while visual guidance becomes more critical. Our DVPM makes the foremost impact here by enhancing visual guidance richness through rank information prior integration. Its removal (fourth experimental row) leads to 5.52% mIoU and 3.36% FB-IoU reductions. Meanwhile, the RFFM module improves performance moderately through cross-modal prior-based fusion of reliable query and support features.

TABLE VI

COMPARATIVE RESULTS WITH OTHER CROSS-MODAL PRIORS ON iSAID-$5^i$, EVALUATED USING MIOU (%) AND FB-IOU (%). THE BASELINE IS THE BASE MODEL THAT RETAINS ONLY COSINE SIMILARITY, AND "W/" DENOTES COMBINATIONS OF THE BASELINE WITH DIFFERENT PRIORS

| Setting | Methods | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU |
|---|---|---|---|---|---|---|
| 1-shot | Baseline | 39.62 | 25.83 | 38.62 | 34.69 | 57.39 |
| | w/ Cosine Similarity | 42.02 | 26.05 | 45.60 | 37.89 | 57.64 |
| | w/ MaskCLIP [7] | 35.00 | 24.39 | 33.84 | 31.08 | 54.63 |
| | w/ GEM [8] | 36.36 | 25.94 | 38.10 | 33.47 | 55.43 |
| | w/ Softmax-GradCAM [6] | 42.99 | 30.90 | 46.84 | 40.24 | 59.03 |
| | w/ IPDM (Ours) | **52.79** | **34.20** | **48.86** | **45.28** | **61.38** |
| 5-shot | Baseline | 46.37 | 32.09 | 47.93 | 42.13 | 61.05 |
| | w/ Cosine Similarity | 46.95 | 30.17 | 49.70 | 42.27 | 60.02 |
| | w/ MaskCLIP [7] | 42.70 | 28.81 | 44.39 | 38.63 | 58.62 |
| | w/ GEM [8] | 44.00 | 30.00 | 46.37 | 40.12 | 59.21 |
| | w/ Softmax-GradCAM [6] | 48.28 | 34.22 | 50.62 | 44.47 | 61.0 |
| | w/ IPDM (Ours) | **55.39** | **36.25** | **53.49** | **48.38** | **64.24** |

*2) Cross-Modal Prior Comparisons:* To validate and analyze the effectiveness of our IPDM, we conducted comprehensive comparisons with alternative cross-modal priors, including the undecoupled cosine similarity derived from (3), the Softmax-GradCAM [6], the MaskCLIP [7], and the GEM [8]. To ensure a fair comparison as much as possible, we minimized the influence of other modules. Specifically, we removed all modules from our network and retained only the cosine similarity prior as the baseline. Subsequently, we separately introduced different cross-modal priors to conduct experiments on iSAID-$5^i$, with comparative results detailed in Table VI. Our method demonstrates superior performance over both alternatives across all one-shot/five-shot settings and different folds. Furthermore, Fig. 1 reveals critical insights: 1) the direct application of cosine similarity from (3) yields ambiguous and underutilized priors due to CLIP's lack of dense prediction task specialization; 2) while Softmax-GradCAM proves effective in natural image dense task, its performance degrades significantly on complex remote sensing imagery; and 3) methods such as MaskCLIP and GEM, which achieve visual semantic extraction through extensive modifications to the CLIP architecture, fail in remote sensing scenarios, and this may be caused by differences between models. In contrast, our IPDM strategically incorporates background textual prompts to enhance target-class prior precision, thereby significantly improving cross-modal prior utility. This improvement is quantitatively evidenced by the substantial performance gaps shown in Table VI, confirming IPDM's critical role in remote sensing FSS.

*3) Analysis of Shuffle Operation in IPDM:* As formalized in (5), we apply random shuffling to background vision–language

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TABLE VII

ABLATION STUDY ON THE SHUFFLE OPERATION IN IPDM FOR iSAID-$5^i$, EVALUATED USING MIOU (%) AND FB-IOU (%). GREEN RESULTS INDICATE THE EXTENT OF PERFORMANCE DEGRADATION, WHILE RED RESULTS REFLECT THE MAGNITUDE OF PERFORMANCE ENHANCEMENT

| Shot | Shuffle | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU |
|---|---|---|---|---|---|---|
| 1-shot | with | **58.63** | **38.76** | **56.82** | **51.40** | **66.56** |
| | w/o | 55.51 | 34.85 | 41.79 | 44.05 | 61.36 |
| | | −3.12 | −3.91 | −15.03 | −7.35 | −5.20 |
| 5-shot | with | 60.09 | **43.46** | **58.15** | **53.90** | **68.73** |
| | w/o | **60.27** | 40.71 | 50.57 | 50.52 | 65.93 |
| | | +0.18 | −2.75 | −7.58 | −3.38 | −2.80 |

TABLE VIII

ABLATION STUDY ON THE LOSS OF THE PRIOR LOSS $\mathcal{L}_s^{\text{CLIP}}$ IN IPDM FOR iSAID-$5^i$ IN ONE-SHOT SETTING, EVALUATED USING MIOU (%) AND FB-IOU (%). GREEN RESULTS INDICATE THE EXTENT OF PERFORMANCE DEGRADATION

| $\mathcal{L}_s^{clip}$ | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU |
|---|---|---|---|---|---|
| with | **58.63** | **38.76** | **56.82** | **51.40** | **66.56** |
| w/o | 57.17 | 35.74 | 51.46 | 48.12 | 63.68 |
| | −1.46 | −3.02 | −5.36 | −3.28 | −2.88 |

prompts to prevent the introduction of explicit categorical information about background classes. This approach effectively mitigates potential distributional bias between base and novel datasets caused by fixed class ordering in textual prompts. To validate this design rationale, we conducted an ablation study on the shuffling operation, with results summarized in Table VII. The absence of shuffling leads to severe performance degradation, most notably in the one-shot scenario of fold-2, where mIoU drops by 15.03%. This empirical evidence confirms that our shuffled prompt strategy successfully eliminates dataset-specific prior biases while maintaining the discriminative power of cross-modal representations.

*4) Analysis of Prior Loss in IPDM:* To further leverage the information provided by support images to enhance the prior decoupling capability of the prior learner, we use the cross-entropy between support labels and the prior as the prior loss $\mathcal{L}_s^{\text{clip}}$. To evaluate the effect of this additional loss on the model, we conducted ablation experiments under the one-shot setting on the iSAID-$5^i$ dataset, with results shown in Table VIII. The model without this loss exhibited performance drops across all three folds, particularly in fold-2, where the mIoU decreased by as high as 5.36%. Overall, the average mIoU across the three folds dropped by 3.28%, and the FB-IoU decreased by 2.88%. These results validate the effectiveness of the prior loss for IPDM.

*5) Analysis of Various Priors:* As illustrated in Fig. 6, we visualize the cross-modal prior and multilevel hierarchical priors of the model on the iSAID-$5^i$ dataset under a one-shot setting with ResNet-50 as the backbone, aiming to analyze their contributions to segmentation results. Fig. 6(a) displays our proposed cross-modal prior, while (b)–(g) presents the cosine similarity prior and rank information prior derived from the final layers of the second, third, and fourth stages of ResNet-50, respectively. Observations reveal

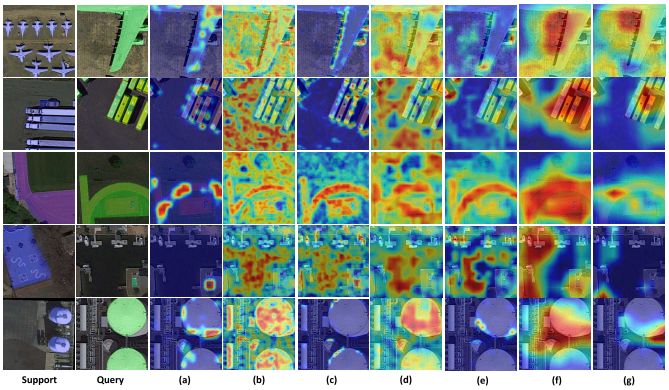



Fig. 6. Visualization results of heterogeneous priors on iSAID-$5^i$. (a) Cross-modal prior. (b)–(g) Cosine similarity prior and rank information prior from the final layers of stages 2–4 in ResNet-50, respectively.

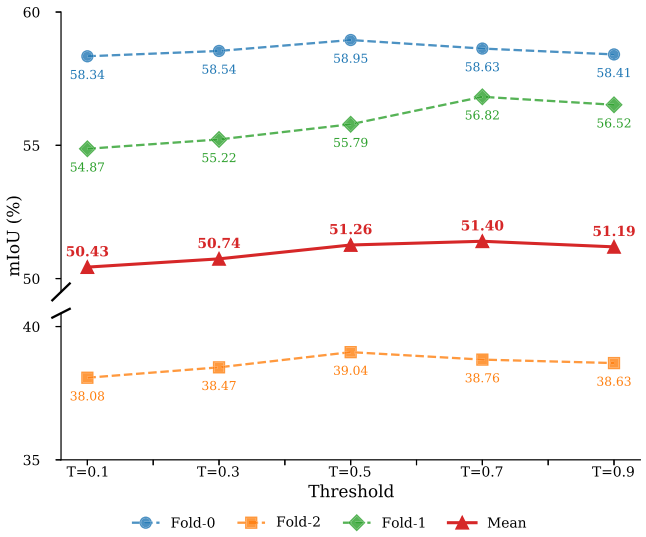



Fig. 7. Analysis of threshold $\tau$ variations for RFFM in one-shot setting of iSAID-$5^i$.

that the CLIP prior exhibits concentrated high-confidence regions with relatively superior precision, enabling it to play an irreplaceable role in scenarios with significant intraclass variations, as seen in the third row of Fig. 6(a). However, due to its limited spatial coverage, it underperforms in recalling target-class pixels, ultimately serving as an auxiliary component in FSS. From Fig. 6(b) to (g), our proposed rank information prior demonstrates robust performance under large intraclass variations, particularly in middle-to-lower layers [e.g., rows 1 and 2 in Fig. 6(c) and (e)]. In contrast, the cosine similarity prior complements the rank information prior by enhancing the recall rate of predictions. This synergy highlights the effectiveness of integrating complementary priors to balance precision and recall in challenging segmentation tasks.

*6) Analysis of the Threshold in RFFM:* The prior threshold $\tau$ in RFFM critically influences the quality of the mixed support features. A threshold that is too low introduces excessive erroneous query features, while a threshold that is too high restricts the diversity of query features. Thus, identifying the optimal threshold is essential. To investigate

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TABLE IX
ABLATION STUDY OF THE RANKING METRIC OF RANK INFORMATION PRIOR ON iSAID-$5^i$, EVALUATED USING MIOU (%) AND FB-IOU (%)

| Setting | Prior Type | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU |
|---|---|---|---|---|---|---|
| 1-shot | HR | 56.25 | 37.58 | 50.71 | 48.18 | 64.28 |
| | MRR | 54.83 | 37.20 | 50.59 | 47.54 | 63.16 |
| | NDCG | **58.63** | **38.76** | **56.82** | **51.40** | **66.56** |
| 5-shot | HR | 59.07 | 42.31 | 54.65 | 52.01 | 67.11 |
| | MRR | 58.86 | 41.95 | 56.51 | 52.44 | 67.05 |
| | NDCG | **60.09** | **43.46** | **58.15** | **53.90** | **68.73** |

this, we conducted parameter experiments on $\tau$ using one-shot tasks on the iSAID-$5^i$ dataset, sampling $\tau$ values from [0.1, 0.3, 0.5, 0.7, 0.9]. The detailed experimental results, shown in Fig. 7, demonstrate that the performance across all Folds aligns with the previously mentioned trend. The highest average mIoU across the three folds is achieved when $\tau$ is set to 0.7. Therefore, our model selects 0.7 as the final threshold.

*7) Analysis of Ranking Metric Methods:* Besides the NDCG, ranking metrics such as hit ratio (HR) and mean reciprocal rank (MRR) [66] are also widely used. However, in the proposed rank information prior, NDCG theoretically demonstrates superior properties. Unlike HR, which lacks sensitivity to positional differences, and MRR, which focuses solely on the ranking of the first relevant foreground supporting feature, NDCG not only emphasizes the initial relevant feature but also incorporates subsequent relevant features. It comprehensively computes the prior weights based on the positional significance of all relevant features in the ranking list. We conducted ablation studies by replacing the NDCG-based rank information prior with HR- and MRR-based alternatives. The experimental results in Table IX demonstrate that NDCG achieves superior performance, validating our theoretical analysis.

### C. Limitations Discussions

Although our method demonstrates excellent performance, we must acknowledge that it has certain limitations. First, our approach requires the introduction of additional VLMs, which increases GPU memory demands and leads to longer training and inference times, although the weights of the VLMs are frozen. Second, the effectiveness of our method depends on the quality of the VLMs employed. Moreover, in utilizing VLMs, we discard the [CLS] token, which may contain crucial class information. Investigating the role and underlying mechanisms of the [CLS] token in FSS remains an important direction for future research.

## VI. CONCLUSION

In this article, we introduce the ICPD-Net, a novel approach designed to address intraclass variance in remote sensing FSS by utilizing cross-modal priors derived from vision–language VLMs and incorporating ranking information to enhance visual metric methodologies. Initially, we present the IPDM, which generates class-agnostic vision–language prompts by exploiting ambiguous similarities between target and background classes. This module employs a prior learner that disentangles textual prompts, thereby producing instructive cross-modal priors that aim to overcome the limitations of existing methodologies in remote sensing applications. Additionally, to mitigate intraclass variance, we propose the RFFM, which integrates high-confidence query features with support features through the use of cross-modal priors, thereby enhancing the effectiveness of relational learning via a self-support strategy. Furthermore, we develop a DVPM that amalgamates traditional cosine similarity priors with an innovative rank information prior to enhance visual metric performance. Comprehensive experiments conducted on the iSAID-$5^i$ and DLRSD-$5^i$ datasets substantiate the efficacy of the ICPD-Net, showcasing state-of-the-art performance with significant improvements. The ICPD-Net represents a pioneering and effective framework for the application of VLMs in dense prediction tasks within the realm of remote sensing, thereby contributing to the advancement of FSS.

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